5. Explain briefly seven types of unit cell.
6. Distinguish between hexagonal close packing and cubic close packing.
7. Distinguish tetrahedral and octahedral voids.
8. What are point defects?
9. Explain Schottky defect.
10. Write short note on metal excess and metal deficiency defect with an example.
11. Calculate the number of atoms in a fcc unit cell.
12. Explain AAAA and ABABA and ABCABC type of three dimensional packing with the help of neat diagram.
13. Why ionic crystals are hard and brittle?
14. Calculate the percentage efficiency of packing in case of body centered cubic crystal.
15. What is the two dimensional coordination number of a molecule in square close packed layer?
16. Experiment shows that Nickel oxide has the formula $Ni_{0.96}O_{1.00}$. What fraction of Nickel exists as of Ni^{2+} and Ni^{3+} ions?
17. What is meant by the term "coordination number"? What is the coordination number of atoms in a bcc structure?
18. An element has bcc structure with a cell edge of 288 pm. the density of the element is 7.2 gcm^{-3}. how many atoms are present in 208g of the element.
19. Aluminium crystallizes in a cubic close packed structure. Its metallic radius is 125pm. calculate the edge length of unit cell.
20. if NaCl is doped with 10^{-2} mol percentage of strontium chloride, what is the concentration of cation vacancy?
21. KF crystallizes in fcc structure like sodium chloride. calculate the distance between K^+ and F^- in KF.(given : density of KF is $2.48\ g\ cm^{-3}$)
22. An atom crystallizes in fcc crystal lattice and has a density of $10\ gcm^{-3}$ with unit cell edge length of 100pm. calculate the number of atoms present in 1 g of crystal.
23. Atoms X and Y form bcc crystalline structure. Atom X is present at the corners of the cube and Y is at the centre of the cube. What is the formula of the compound?
24. Sodium metal crystallizes in bcc structure with the edge length of the unit cell $4.3\times10^{-8}\ cm$. calculate the radius of sodium atom.
25. Write a note on Frenkel defect.

XII U6 Solid State - Jerald.indd 201 3/29/2019 6:30:39 PM